

AML Assignment 1

Tender Laura - Maria
Group 411 Data Science

April 26, 2023

1. (0.75 points) Give examples for:

- a) a finite hypothesis class $\mathcal{H}$ with $\text{VCdim}(\mathcal{H}) = 2023$. Justify your choice. (0.25 points)
- b) an infinite hypothesis class $\mathcal{H}$ with $\text{VCdim}(\mathcal{H}) = 2023$. Justify your choice. (0.25 points)
- c) an infinite hypothesis class $\mathcal{H}$ with $\text{VCdim}(\mathcal{H}) = \infty$. Justify your choice. (0.25 points)

Solution:

- b) We consider $\mathcal{H} = \mathcal{HS}_0^n$ the set of halfspaces (linear classifiers) in $\mathbb{R}^n$ that go through origin.

$$\mathcal{HS}_0^n = \{h_{w,0} : \mathbb{R}^n \rightarrow \{-1, 1\}, h_{w,0}(x) = \text{sign}\left(\sum_{i=1}^n w_i x_i\right) \mid w \in \mathbb{R}^n\}$$

In Lecture 7, we proved that $\text{VCdim}(\mathcal{HS}_0^n) = n$. So, as a result for this exercise will chose $\mathcal{HS}_0^{2023}$ as $\text{VCdim}(\mathcal{HS}_0^{2023}) = 2023$.

$$\mathcal{HS}_0^{2023} = \{h_{w,0} : \mathbb{R}^{2023} \rightarrow \{-1, 1\}, h_{w,0}(x) = \text{sign}\left(\sum_{i=1}^{2023} w_i x_i\right) \mid w \in \mathbb{R}^{2023}\}$$

which is an infinite hypothesis class.

- a) Using the previous subtask we'll take $\mathcal{H}$ a subset of $\mathcal{HS}_0^{2023}$ having $|\mathcal{H}| = 2023$.

$$\mathcal{H} = \{h_{w,0} : \mathbb{R}^{2023} \rightarrow \{-1, 1\}, h_{w,0}(x) = \text{sign}\left(\sum_{i=1}^{2023} w_i x_i\right) \mid w \in A\},$$

where $A \subset \mathbb{R}^{2023}$, $A = \{(w_1, w_2, \dots, w_{2023}) \mid w_i = 1, w_j = 0, j \neq i, i = \overline{1, 2023}\}$, which is equivalent to

$$\mathcal{H} = \{h_{i,0} : \mathbb{R}^{2023} \rightarrow \{-1, 1\}, h_{i,0}(x) = \text{sign}(x_i) \mid i = \overline{1, 2023}\},$$

a finite hypothesis class and $\text{VCdim}(\mathcal{H}) = 2023$.

- c) As presented in Lecture 7, the hypothesis class $\mathcal{H}_{\sin} = \{h_\theta : \mathbb{R} \rightarrow \{0, 1\} \mid h_\theta(x) = [\sin(\theta x)], \theta \in \mathbb{R}\}$, $[-1] = 0$ is infinite and has $\text{VCdim}(\mathcal{H}) = \infty$. We proved that n points, for any $n \in \mathbb{N}$ can be shattered by $\mathcal{H}_{\sin}$.

2. (0.75 points) Consider $\mathcal{H}$ to be the following hypothesis class:

$$\mathcal{H} = \{h_a : \mathbb{R}^3 \rightarrow \{0, 1\} \mid h_a(\mathbf{x}) = \mathbb{1}_{\|\mathbf{x}\|_2 \leq a}(\mathbf{x}), \mathbf{x} = (x_1, x_2, x_3) \in \mathbb{R}^3, \|\mathbf{x}\|_2 = \sqrt{x_1^2 + x_2^2 + x_3^2}\}$$

Consider the realizability assumption.

- show that $\mathcal{H}$ can be (ϵ, δ) -PAC learned by giving an algorithm A and determining the sample complexity $m_{\mathcal{H}}(\epsilon, \delta)$ such that the definition of PAC learnability is satisfied. (0.5 points)
- compute $\text{VCdim}(\mathcal{H})$. (0.25 points)

Solution:

Let's have a look at the general equation of a sphere:

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = r^2,$$

where (a, b, c) represents the center of the sphere and r represents the radius. So a sphere centered in the origin $(0, 0, 0)$ will have the equation $x^2 + y^2 + z^2 = r^2$. We can observe that our problem translates to

$$\mathcal{H} = \{h_a : \mathbb{R}^3 \rightarrow \{0, 1\} \mid h_a(\mathbf{x}) = 1 \text{ if } \mathbf{x} \in S(O, a) \text{ otherwise } 0, \mathbf{x} = (x_1, x_2, x_3)\},$$

where $S(O, a)$ is a sphere centered in the origin with radius a .

- We'll prove that $\text{VCdim}(\mathcal{H}) = 1$ in two steps: $\text{VCdim}(\mathcal{H}) \geq 1$ and $\text{VCdim}(\mathcal{H}) < 2$.

First step: Let's choose a point $x = (x_1, x_2, x_3) \in \mathbb{R}^3$ having $\|x\|_2 = \sqrt{x_1^2 + x_2^2 + x_3^2} > 0$. To get label 1 we'll choose $a \geq \|x\|_2$. To get label 0 we'll choose $a < \|x\|_2$.

Second step: Let's suppose there are 2 points that are shattered by $\mathcal{H}$: $x_1, x_2 \in \mathbb{R}^3, \|x_1\|_2 \leq \|x_2\|_2$. We can't get the label $(0, 1)$ meaning that the first point (which is closer to the origin) is not in the sphere but the second one is.

To conclude, we proved $\text{VCdim}(\mathcal{H}) = 1$.

- From the definition of PAC-learnability, we know that $\mathcal{H}$ is PAC-learnable if there exists a function $m_{\mathcal{H}} : (0, 1)^2 \rightarrow \mathbb{N}$ and there exists a learning algorithm A with the following property: for every $\epsilon, \delta > 0$, for every labeling function $f \in \mathcal{H}$ (realizability case), for every distribution $\mathcal{D}$ on $\mathbb{R}$ when we run the learning algorithm A on a training set S consisting of $m \geq m_{\mathcal{H}}(\epsilon, \delta)$ examples sampled i.i.d. from $\mathbb{D}$ and labeled by f , the algorithm A returns a hypothesis $h_s \in \mathcal{H}$ such that, with probability at least $1 - \delta$ (over the choice of examples), the real risk of h_s is smaller than ϵ :

$$\begin{aligned} P_{S \sim \mathcal{D}^m} (L_{f, \mathcal{D}}(h_S) \leq \epsilon) &\geq 1 - \delta \text{ or otherwise said} \\ P_{S \sim \mathcal{D}^m} (L_{f, \mathcal{D}}(h_S) > \epsilon) &< \delta \end{aligned}$$

First, we need to find the algorithm A .

We are under the realizability assumption, so there exists a labeling function $f \in \mathcal{H}, f = h_{(a^*)}$ that labels the training data.

$$\text{Consider the training set } S = \left\{ (x_1, y_1), (x_2, y_2), \dots, (x_m, y_m) \mid \begin{array}{l} y_i = h_{(a^*)}(x_i), \\ x_i \in \mathbb{R}^3, x_i = (x_{i1}, x_{i2}, x_{i3}) \end{array} \right\}$$

$h_S = h_{(a_S)}$, where

$$a_S = \max_{\substack{i=1,m \\ y_i=1}} \sqrt{x_{i1}^2 + x_{i2}^2 + x_{i3}^2}$$

We are under the realizability assumption, so $(0,0,0)$ will never be labeled as one. In case S has no point labeled as 1, A will return $a_S = 0$.

By construction, A is an ERM, meaning that $L_{h^*, \mathcal{D}}(h_S) = 0$, h_S doesn't make any errors on the training set S .

Now we want to find the sample complexity $m_{\mathcal{H}}(\epsilon, \delta)$ such that

$$\mathbb{P}_{S \sim \mathcal{D}^m} (L_{h^*, \mathcal{D}}(h_S) \leq \epsilon) \geq 1 - \delta \text{ where } S \text{ contains } m \geq m_{\mathcal{H}}(\epsilon, \delta) \text{ examples.}$$

Let S_S be the smallest sphere enclosing all positive examples and S^* the sphere that generates the labels. We make the observation that h_S makes errors in region $S^* \setminus S_S$, assigning the label 0 to points that should get label 1. All points $\in S_S$ will be labeled correctly (label 1), all points outside S^* will be labeled correctly (label 0).

Let's fix $\epsilon > 0, \delta > 0$ and consider a distribution $\mathcal{D}$ over $\mathbb{R}^2$.

Case 1)

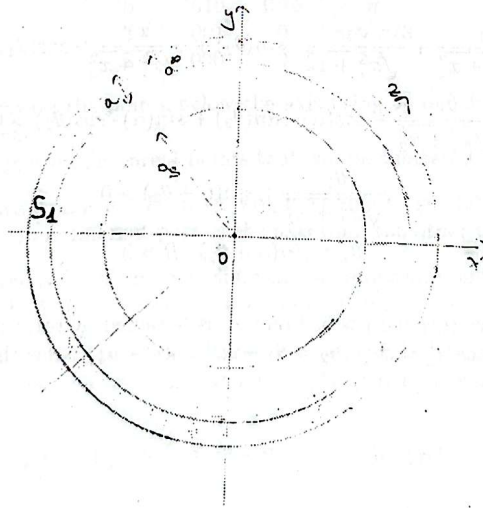
If $\mathcal{D}(S^*) = \mathbb{P}_{x \sim \mathcal{D}}(x \in S^*) \leq \epsilon$ then in this case

$$L_{h^*, \mathcal{D}}(h_S) = \mathbb{P}_{x \sim \mathcal{D}}(h_S(x) \neq h^*(x)) = \mathbb{P}_{x \sim \mathcal{D}}(x \in S^* \setminus S_S) \leq \mathbb{P}_{x \sim \mathcal{D}}(x \in S^*) \leq \epsilon \text{ so we have that}$$

$$\mathbb{P}_{S \sim \mathcal{D}^m} (L_{h^*, \mathcal{D}}(h_S) \leq \epsilon) = 1 \text{ (this happens all the time)}$$

Case 2) $\mathcal{D}(R^*) = \mathbb{P}_{x \sim \mathcal{D}}(x \in R^*) > \epsilon$

We construct a geometric shape S_1 with $\mathcal{D}(S_1) = \mathbb{P}_{x \sim \mathcal{D}}(x \in S_1) = \epsilon$. (See the picture below.)



If S_S (the sphere returned by A , implemented by h_S) intersects S_1 :

$$L_{h^*, \mathcal{D}}(h_S) = P_{x \sim \mathcal{D}}(h^*(x) \neq h_S(x)) = P_{x \sim \mathcal{D}}(x \in S^* \setminus S_S) \leq P_{x \sim \mathcal{D}}(x \in S_1) = \epsilon$$

So, in this case, $P_{S \sim \mathcal{D}^m}(L_{h^*, \mathcal{D}}(h_S) \leq \epsilon) = 1$ (this happens always).

In order to have $L_{h^*, \mathcal{D}}(h_S) > \epsilon$, we need that S_S will not intersect S_1 . We denote with F this event, so we have $F = \{S \sim \mathcal{D}^m \mid S_S \cap S_1 = \emptyset\}$. This leads to the following:

$$P_{S \sim \mathcal{D}^m}(L_{h^*, \mathcal{D}}(h_S) > \epsilon) \leq P_{S \sim \mathcal{D}^m}(F) = (1 - \epsilon)^m \leq e^{-\epsilon m}$$

We want to make this probability smaller than δ :

$$e^{-\epsilon m} < \delta \mid \log_e$$

$$-\epsilon \cdot m < \log \delta$$

$$m > -\frac{1}{\epsilon} \log \delta = \frac{1}{\epsilon} \log \frac{1}{\delta}$$

So, we take $m \geq m_{\mathcal{H}}(\epsilon, \delta) = \frac{1}{\epsilon} \log \frac{1}{\delta}$, to satisfy PAC learnability.

3. (1 point) Compute $\text{VCdim}(\mathcal{H})$ where $\mathcal{H}$ is the following hypothesis class:

$$\mathcal{H} = \{h_{\theta_1, \theta_2} : \mathbb{R}^2 \rightarrow \{0, 1\} \mid h_{\theta_1, \theta_2}(\mathbf{x}) = h_{\theta_1, \theta_2}(x_1, x_2) = \mathbb{1}_{[\theta_1 + x_1 \cdot \sin(\theta_2) + x_2 \cdot \cos(\theta_2) > 0]}, \theta_1, \theta_2 \in \mathbb{R}, \mathbf{x} \in \mathbb{R}^2\}.$$

Solution:

Let's consider our point $A(x_1, x_2)$, a point with radius $R = \sqrt{x_1^2 + x_2^2}$ at an angle α and have a look at the meaning of the inequation:

$$\theta_1 + x_1 \cdot \sin(\theta_2) + x_2 \cdot \cos(\theta_2) > 0 \mid : \sqrt{x_1^2 + x_2^2}$$

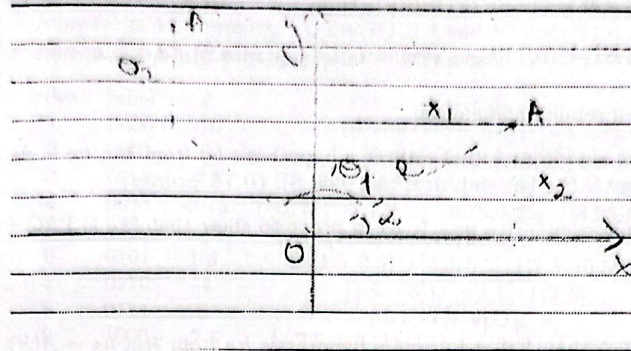
$$\frac{\theta_1}{\sqrt{x_1^2 + x_2^2}} + \frac{x_1}{\sqrt{x_1^2 + x_2^2}} \cdot \sin(\theta_2) + \frac{x_2}{\sqrt{x_1^2 + x_2^2}} \cdot \cos(\theta_2) > 0$$

$$\frac{\theta_1}{\sqrt{x_1^2 + x_2^2}} + \cos(\alpha) \cdot \sin(\theta_2) + \sin(\alpha) \cdot \cos(\theta_2) > 0$$

$$\frac{\theta_1}{\sqrt{x_1^2 + x_2^2}} + \sin(\alpha + \theta_2) > 0$$

$$\theta_1 + \sin(\alpha + \theta_2) \cdot R > 0$$

Point A would be above (Ox if $x_2 > 0$). Point A' is obtained by rotating point A to the left with θ_2 and ascended with θ_1 has $x'_2 = x_2 + \theta_2 = \theta_2 + R \cdot \sin(\alpha + \theta_1)$. (See the picture below.)



So, we can translate our problem in natural language by saying if the point obtained by rotating with θ_2 and raising with θ_1 the previous point x is above the Ox axis then $h_{\theta_1, \theta_2}(x)$ equals 1 otherwise 0.

We'll prove that $\text{VCdim}(\mathcal{H}) = 3$ in two steps: $\text{VCdim}(\mathcal{H}) \geq 3$ and $\text{VCdim}(\mathcal{H}) < 4$.

First step:

Let's pick the following three points: $(R_3 = 1, \alpha_3 = \frac{2\pi}{3}), (R_2 = 0.5, \alpha_2 = \frac{\pi}{2}), (R_1 = 1, \alpha_1 = \frac{\pi}{6})$. Now let's pick θ_1, θ_2 in order to get all 8 labels.

index	label	θ_1	θ_2
1	111	0	0
2	110	0	$-\pi/3$
3	101	-0.75	0
4	100	0	$-\pi/2$
5	011	0	$\pi/3$
6	010	0.75	π
7	001	0	$\pi - \pi/3$
8	000	0	$+\pi$

As a summary I either rotate the points below the axis using θ_2 or I lower them by decreasing θ_1 and $\mathcal{H}$ can shatter 3 points.

Second step: Let's suppose there are 4 points that can be shattered by $\mathcal{H}$.

- 1) At least 3 points are collinear. Then we won't be able to get label 101. We can't have the point in the middle on a different part of the axis than the other two.
- 2) The 4 points form a convex polygon which means two adjacent points will be above the polygon center and two will be below. So we can't get label 1010.
- 3) The 4 points form a concave polygon which means one of the points will be inside the triangle form by the other three. So we can't get label 0100 (where 1 is assigned to the point inside the triangle).

In no situation $\mathcal{H}$ can shatter 4 points. To conclude, $\text{VCdim}(\mathcal{H}) \geq 3$ and $\text{VCdim}(\mathcal{H}) < 4$, so

$$\text{VCdim}(\mathcal{H}) = 3.$$

4. (1.5 points) Consider $\mathcal{H}_\alpha$ to be the class of axis aligned rectangles with fixed aspect-ratio α , where $\alpha \in \mathbb{R}$ and $\alpha > 0$:

$$\mathcal{H}_\alpha = \{h_{a,b,c,d} : \mathbb{R}^2 \rightarrow \{0,1\} | h_{a,b,c,d}(x) = \mathbb{1}_{[a,b] \times [c,d]}(x), a,b,c,d \in \mathbb{R}, a < b, c < d, \frac{d-c}{b-a} = \alpha\}$$

Consider the realizability assumption.

- give a learning algorithm A that returns a hypothesis h_S from $\mathcal{H}_\alpha$, $h_S = A(S)$ consistent with the training set S (h_S has empirical risk 0 on S). (0.75 points)
- find the sample complexity $m_{\mathcal{H}_\alpha}(\epsilon, \delta)$ in order to show that $\mathcal{H}_\alpha$ is PAC - learnable. (0.75 points)

Solution:

- For finding algorithm A that returns a hypothesis h_S from $\mathcal{H}_\alpha$, $h_S = A(S)$ we'll consider the realizability assumption meaning all the points in S can be labeled correctly.

$$\text{Consider the training set } S = \left\{ (x_1, y_1), (x_2, y_2), \dots, (x_m, y_m) \mid \begin{array}{l} y_i = h_{(a_1^*, b_1^*, a_2^*, b_2^*)}(x_i), \\ x_i \in \mathbb{R}, x_i = (x_{i1}, x_{i2}) \end{array} \right\}.$$

A will look for the smallest axis aligned rectangle with fixed aspect-ratio α so that all and only the points labeled as 1 are inside it.

Consider the following algorithm A , that takes as input the training set S . Now we'll use the algorithm from the first seminar to find the smallest axis aligned rectangle containing the points.

$$h'_S = h_{(a_{1S}, b_{1S}, a_{2S}, b_{2S})}, \text{ where}$$

$$a_{1S} = \min_{\substack{i=1, \dots, m \\ y_i=1}} x_{i1}$$

$$b_{1S} = \max_{\substack{i=1, \dots, m \\ y_i=1}} x_{i1}$$

$$a_{2S} = \min_{\substack{i=1, \dots, m \\ y_i=1}} x_{i2}$$

$$b_{2S} = \max_{\substack{i=1, \dots, m \\ y_i=1}} x_{i2}$$

Now we'll need to make our rectangle have a fixed aspect-ratio. h_S will be equal to the result of h'_S to which we apply the following rules:

- If $\frac{d_{1S} - c_{1S}}{b_{1S} - a_{1S}} = \alpha$ then we found the answer.
- If $\frac{d_{1S} - c_{1S}}{b_{1S} - a_{1S}} < \alpha$ then we increase d_{1S} either until the ratio equals α either until we encounter the first point labeled negative. In the latter we will decrease c_{1S} to meet the ratio.
- If $\frac{d_{1S} - c_{1S}}{b_{1S} - a_{1S}} > \alpha$ then we increase b_{1S} either until the ratio equals α either until we encounter the first point labeled negative. In the latter we will decrease a_{1S} to meet the ratio.

Edge case, if no label is 1 we'll choose a small rectangle $b - a = \epsilon, d - c = \epsilon \cdot \alpha$ so that no point is inside it.

- (1 point) Compute $\text{VCdim}(\mathcal{H})$ where $\mathcal{H}$ is the following hypothesis class:

$$\mathcal{H} = \{h_\theta : \mathbb{R} \rightarrow \{0,1\} | h_\theta(x) = \mathbb{1}_{[\theta, \theta+1] \cup [\theta+2, \theta+4] \cup [\theta+6, \theta+9]}(x), \theta \in \mathbb{R}\}.$$

Solution:

For 7 points will never be able to get the label 1010101, so an upper bound for $\text{VCdim}(\mathcal{H})$ is 7.

We'll prove $\text{VCdim}(\mathcal{H}) = 4$ by proving $\text{VCdim}(\mathcal{H}) \geq 4$ and $\text{VCdim}(\mathcal{H}) < 5$.

I'll pick the points $\{1.5, 2, 3.5, 6\}$ and find θ in order to get all the 2^4 labels.

index	label	θ	intervals
1	0000	-10	$[-10, -9] \cup [-8, -6] \cup [-4, -1]$
2	0001	6	$[6, 7] \cup [8, 9] \cup [12, 15]$
3	0010	-3.5	$[-3.5, -2.5] \cup [-1.5, 0.5] \cup [2.5, 5.5]$
4	0011	-2.75	$[-2.75, -1.75] \cup [-0.75, 1.25] \cup [3.25, 6.25]$
5	0100	1.75	$[1.75, 2.75] \cup [3.75, 5.75] \cup [7.75, 10.75]$
6	0101	1.5	$[1.5, 2.5] \cup [3.5, 5.5] \cup [7.5, 10.5]$
7	0110	-4	$[-4, -3] \cup [-2, 0] \cup [2, 5]$
8	0111	0	$[0, 1] \cup [2, 4] \cup [6, 9]$
9	1000	-7.5	$[-7.5, -6.5] \cup [-5.5, -3.5] \cup [-1.5, 1.5]$
10	1001	-2.25	$[-2.25, -1.25] \cup [-0.25, 1.75] \cup [3.75, 6.25]$
11	1010	0.5	$[0.5, 1.5] \cup [2.5, 4.5] \cup [6.5, 9.5]$
12	1011	-2.5	$[-2.5, -1.5] \cup [-0.5, 1.5] \cup [3.5, 6.5]$
13	1100	-7	$[-7, -6] \cup [-5, -3] \cup [-1, 2]$
14	1101	-0.75	$[-0.75, 0.25] \cup [1.25, 3.25] \cup [5.25, 8.25]$
15	1110	1	$[1, 2] \cup [3, 5] \cup [7, 10]$
16	1111	-0.5	$[-0.5, 0.5] \cup [1.5, 3.5] \cup [5.5, 8.5]$

We'll assume there are 5 points $x_1 < x_2 < x_3 < x_4 < x_5$ that can give us all 2^{32} labels. Getting the label 10101:

x_1	x_2	x_3	x_4	x_5
1	0	1	0	1
$[\theta, \theta + 1]$	$(\theta + 1, \theta + 2)$	$[\theta + 2, \theta + 4]$	$(\theta + 4, \theta + 6)$	$[\theta + 6, \theta + 9]$
$ \begin{aligned} 0 &< x_2 - x_1 < 2 \\ 0 &< x_3 - x_2 < 3 \\ 0 &< x_4 - x_3 < 4 \\ 0 &< x_5 - x_4 < 5 \\ 1 &\leq x_3 - x_1 \leq 4 \\ 2 &< x_4 - x_2 < 5 \\ 2 &\leq x_5 - x_3 \leq 7 \\ 5 &\leq x_5 - x_1 \leq 9 \end{aligned} $				

Getting the label 10001:

Case 1)

x_1	x_2	x_3	x_4	x_5
1	0	0	0	1
$[\theta, \theta + 1]$	$(\theta + 1, \theta + 2)$	$(\theta + 1, \theta + 2)$	$(\theta + 4, \theta + 6)$	$[\theta + 6, \theta + 9]$

$$\begin{array}{rcl}
0 & < x_2 - x_1 < & 2 \\
0 & < x_3 - x_2 < & 1 \\
2 & < x_4 - x_3 < & 4 \\
0 & < x_5 - x_4 < & 5 \\
1 & \leq x_3 - x_1 \leq & 2 \\
2 & < x_4 - x_2 < & 5 \\
4 & \leq x_5 - x_3 \leq & 7 \\
5 & \leq x_5 - x_1 \leq & 9
\end{array}$$

Case 2)

x_1	x_2	x_3	x_4	x_5
1	0	0	0	1
$[\theta, \theta + 1]$	$(\theta + 1, \theta + 2)$	$(\theta + 4, \theta + 6)$	$(\theta + 4, \theta + 6)$	$[\theta + 6, \theta + 9]$

$$\begin{array}{rcl}
0 & < x_2 - x_1 < & 2 \\
2 & < x_3 - x_2 < & 3 \\
0 & < x_4 - x_3 < & 2 \\
0 & < x_5 - x_4 < & 5 \\
2 & \leq x_3 - x_1 \leq & 4 \\
2 & < x_4 - x_2 < & 5 \\
2 & \leq x_5 - x_3 \leq & 5 \\
5 & \leq x_5 - x_1 \leq & 9
\end{array}$$

Case 3)

x_1	x_2	x_3	x_4	x_5
1	0	0	0	1
$[\theta + 2, \theta + 4]$	$(\theta + 4, \theta + 6)$	$(\theta + 5, \theta + 6)$	$(\theta + 5, \theta + 6)$	$[\theta + 7, \theta + 9]$

$$\begin{array}{rcl}
0 & < x_2 - x_1 < & 2 \\
0 & < x_3 - x_2 < & 3 \\
0 & < x_4 - x_3 < & 4 \\
0 & < x_5 - x_4 < & 5 \\
1 & \leq x_3 - x_1 \leq & 4 \\
2 & < x_4 - x_2 < & 2 \Rightarrow \times
\end{array}$$

We'll move forward with the equations from case 1) and choose label 11001. $x_2 - x_1 < 2 \Rightarrow x_2 < \theta + 3$.

x_1	x_2	x_3	x_4	x_5
1	1	0	0	1
$[\theta, \theta + 1]$	$[\theta, \theta + 1]$	$(\theta + 1, \theta + 2)$	$(\theta + 4, \theta + 6)$	$[\theta + 6, \theta + 9]$

$$\begin{array}{rcl}
0 & < x_2 - x_1 < & 1 \\
0 & < x_3 - x_2 < & 1 \\
2 & < x_4 - x_3 < & 4 \\
0 & < x_5 - x_4 < & 5 \\
1 & \leq x_3 - x_1 \leq & 2 \\
3 & < x_4 - x_2 < & 5 \\
4 & \leq x_5 - x_3 \leq & 7 \\
5 & \leq x_5 - x_1 \leq & 9
\end{array}$$

or

$$\begin{array}{ccccc}
 x_1 & x_2 & x_3 & x_4 & x_5 \\
 1 & 1 & 0 & 0 & 1 \\
 [\theta, \theta + 1] & [\theta + 2, \theta + 3] & (\theta + 4, \theta + 6) & (\theta + 4, \theta + 6) & [\theta + 6, \theta + 9]
 \end{array}$$

$$\begin{array}{l}
 1 \leq x_2 - x_1 < 2 \\
 1 < x_3 - x_2 < 1 \Rightarrow \Leftarrow
 \end{array}$$

We'll move forward with the equations from case 2) and choose label 11001.

$$\begin{array}{ccccc}
 x_1 & x_2 & x_3 & x_4 & x_5 \\
 1 & 1 & 0 & 0 & 1 \\
 [\theta, \theta + 1] & [\theta, \theta + 1] & (\theta + 1, \theta + 2) & (\theta + 4, \theta + 6) & [\theta + 6, \theta + 9]
 \end{array}$$

$$\begin{array}{l}
 0 < x_2 - x_1 < 1 \\
 2 < x_3 - x_2 < 2 \Rightarrow \Leftarrow
 \end{array}$$

$$\begin{array}{ccccc}
 x_1 & x_2 & x_3 & x_4 & x_5 \\
 1 & 1 & 0 & 0 & 1 \\
 [\theta, \theta + 1] & [\theta, \theta + 1] & (\theta + 4, \theta + 6) & (\theta + 4, \theta + 6) & [\theta + 6, \theta + 9]
 \end{array}$$

$$\begin{array}{l}
 0 < x_2 - x_1 < 1 \\
 3 < x_3 - x_2 < 3 \Rightarrow \Leftarrow
 \end{array}$$

The inequations from case 2 no longer hold so we'll continue with the last standing inequations. For label 11101 this is the only option because of $x_2 - x_1 < 1$ and $x_3 - x_2 < 1$.

$$\begin{array}{ccccc}
 x_1 & x_2 & x_3 & x_4 & x_5 \\
 1 & 1 & 1 & 0 & 1 \\
 \theta & (\theta, \theta + 1) & \theta + 1 & (\theta + 4, \theta + 5) & [\theta + 6, \theta + 8]
 \end{array}$$

$$\begin{array}{l}
 0 < x_2 - x_1 < 1 \\
 0 < x_3 - x_2 < 1 \\
 3 < x_4 - x_3 < 4 \\
 1 < x_5 - x_4 < 5 \\
 1 \leq x_3 - x_1 \leq 1 \\
 3 < x_4 - x_2 < 4 \\
 5 \leq x_5 - x_3 \leq 7 \\
 6 \leq x_5 - x_1 \leq 8
 \end{array}$$

$$\begin{array}{ccccc}
 x_1 & x_2 & x_3 & x_4 & x_5 \\
 1 & 1 & 1 & 1 & 1 \\
 \theta & (\theta, \theta + 1) & \theta + 1 & (\theta + 4, \theta + 5) & [\theta + 6, \theta + 8]
 \end{array}$$

For label 11111, $x_3 = \theta + 1 \Rightarrow x_4 \in (\theta + 4, \theta + 5)$ which can never be 1 $\Rightarrow \Leftarrow$.
So, $\text{VCdim}(\mathbb{H}) < 5 \Rightarrow \text{VCdim}(\mathbb{H}) = 4$.